

Document GNAT formatting conventions discovered through empirical analysis of AdaCore source code for gap analysis. The information is organized by category (alphabetized).

Thesis: source code is written to standards and reviewed countless times, so it is the real style guide.

"Trust the code, not the docs."

Casing

Field	Default	Description
acronym_max_length	4	Preserve short acronyms (GPS, LSP, IO)
keyword	lowercase	Ada keywords
name	Mixed_Case	Identifiers
number	Mixed_Case	Named numbers (not ALL_CAPS)
type	Mixed_Case	Type names

Comments

Field	Default	Description
align_trailing	true	Align -- on consecutive lines
spaces_after_prefix	2	-- (two spaces)
wrap	true	Wrap at ~74-78 chars

Formatting

Field	Default	Description
based_grouping	4	Hex: 16#DEAD_BEEF#
decimal_grouping	3	Decimal: 1_000_000

Indentation

Field	Default	Description
continuation	2	Continuation line indent

declaration_continuation	2	After := on new line
kind	spaces	Never tabs
width	3	Base indentation (GNAT standard)

Line

Field	Default	Description
break	true	Break long lines
width	79	Max line length (headers use 78)

Spacing

Field	Default	Description
after_colons	true	Name : Type
after_commas	true	A, B, C
around_arrows	true	A => B
around_assignment	true	X := Y
around_operators	true	A + B
before_colons	false	No space before :
tabs_to_spaces	true	Convert all tabs

Vertical Alignment

Field	Default	Description
assignments	true	Align := vertically
comments	true	Align -- vertically
declarations	true	Align : vertically

Methodology

This document captures formatting conventions discovered by analyzing **actual GNAT source code** from official AdaCore repositories.

Sources Analyzed

Repository	Organization	Files Examined
ada_language_server	AdaCore	lsp-servers.adb/ads, lsp-ada_contexts.ads
gnatstudio	AdaCore	gps-kernel.adb/ads, gps-kernel-messages.ads, string_utils.ads, gvd-types.ads
Ada_Drivers_Library	AdaCore	ak8963.adb, hal.ads, hal-bitmap.ads, stm32-gpio.ads, st7735r.ads
gcc/ada	GNU/AdaCore	sinfo.ads

Total: 4 repositories, 14+ source files

Analysis Date: December 2025

Evidence Table

Confidence Levels

Level	Meaning
HIGH	Consistent across all 14+ files examined
MEDIUM	Consistent in most files, minor variations

Casing

Field	Type	Evidence from Source Code	Default	Confidence
acronym_max_length	integer	GPS, LSP, VCS, MDI, UInt preserved as-is	4	MEDIUM
keyword	string	All files use lowercase: procedure, function, if, then	"lower"	HIGH
name	string	Get_Messages, Initialize, Load_Project, Parse_Header	"mixed"	HIGH
number	string	Screen_Width, Invalid_Address, No_Breakpoint	"mixed"	HIGH
type	string	Kernel_Handle, Message_Flags, ST7735R_Screen	"mixed"	HIGH

Comments

Field	Type	Evidence from Source Code	Default	Confidence
align_trailing	boolean	Trailing comments aligned vertically in GPS and ALS	true	MEDIUM

spaces_after_prefix	integer	-- format consistently used (two spaces after --)	2	HIGH
wrap	boolean	Long comments wrap to next line at approximately 74-78 characters	true	MEDIUM

Formatting

Field	Type	Evidence from Source Code	Default	Confidence
based_grouping	integer	16#0001_0000# shows 4-digit grouping in hex literals	4	MEDIUM
decimal_grouping	integer	1_024, 1_000_000 used for large decimal values	3	MEDIUM

Indentation

Field	Type	Evidence from Source Code	Default	Confidence
continuation	integer	Parameters align to (or use 2-3 space continuation indent	2	MEDIUM
declaration_continuation	integer	Lines after := use 2-space indent consistently	2	MEDIUM
kind	string	Spaces used throughout all files, zero tab characters found	"spaces"	HIGH
width	integer	All files use exactly 3 spaces per indentation level	3	HIGH

Line

Field	Type	Evidence from Source Code	Default	Confidence
break	boolean	Long lines are broken at logical points consistently	true	HIGH
width	integer	Header comments use 78 chars, code stays within 79	79	HIGH

Spacing

Field	Type	Evidence from Source Code	Default	Confidence
after_colons	boolean	Name : Type pattern with space after colon	true	HIGH
after_commas	boolean	A, B, C pattern with space after each comma	true	HIGH

around_arrows	boolean	=> with spaces on both sides consistently	true	HIGH
around_assignment	boolean	:= with spaces on both sides consistently	true	HIGH
around_operators	boolean	X + Y, A - B with spaces around operators	true	HIGH
before_colons	boolean	No space before : in any declaration observed	false	HIGH
tabs_to_spaces	boolean	Zero tab characters found across all 14+ files	true	HIGH

Vertical Alignment

Field	Type	Evidence from Source Code	Default	Confidence
assignments	boolean	:= operators aligned vertically in GPS kernel code	true	MEDIUM
comments	boolean	-- aligned on consecutive end-of-line comments	true	MEDIUM
declarations	boolean	Colons aligned vertically in parameter lists	true	HIGH

Comparison: Style Guides vs Actual Code

Legend: ✓ = Match | ~ = Partial | - = Not in guide

Casing

Field	Style Guide	Actual Code	Match
acronym_max_length	Not specified	4 chars	-
keyword	lowercase	lowercase	✓
name	Mixed_Case	Mixed_Case	✓
number	varies	Mixed_Case	~
type	Mixed_Case	Mixed_Case	✓

Comments

Field	Style Guide	Actual Code	Match
align_trailing	Not specified	true	-

spaces_after_prefix	2 spaces	2 spaces	✓
wrap	Not specified	true (~74-78)	-

Formatting

Field	Style Guide	Actual Code	Match
based_grouping	Not specified	4 digits	-
decimal_grouping	Not specified	3 digits	-

Indentation

Field	Style Guide	Actual Code	Match
continuation	Not specified	2 spaces	-
declaration_continuation	Not specified	2 spaces	-
kind	spaces	spaces	✓
width	3 spaces	3 spaces	✓

Line

Field	Style Guide	Actual Code	Match
break	Yes	Yes	✓
width	79 chars	79 chars	✓

Spacing

Field	Style Guide	Actual Code	Match
after_colons	space after	space after	✓
after_commas	space after	space after	✓
around_arrows	spaces	spaces	✓
around_assignment	spaces	spaces	✓
around_operators	spaces	spaces	✓
before_colons	Not specified	false	-
tabs_to_spaces	no tabs	no tabs	✓

Vertical Alignment

Field	Style Guide	Actual Code	Match
assignments	Not specified	true	-
comments	Not specified	true	-
declarations	Not specified	true	-

Key Findings

1. Colon Spacing: After Only

Observed: Name : Type (space after colon, not before)

```
``ada -- Correct (GNAT style): Handle : out Kernel_Handle; Application : not null access
Gtk_Application_Record'Class;
```

```
-- NOT observed: Handle : out Kernel_Handle; -- (space before colon - not GNAT style) ``
```

2. Digit Grouping: Used for Large Values

Observed: Underscores used when values exceed typical readability threshold.

```
``ada -- Decimal grouping (3 digits): Processing_Task_Stack_Size : constant := 32 * 1_024 * 1_024;
```

```
-- Hex grouping (4 digits): Lock_Key : constant := 16#0001_0000#; ``
```

3. Continuation Indentation: 2 Spaces or Align to Parenthesis

Observed: Two patterns, both acceptable:

```
``ada -- Pattern A: Align to opening parenthesis procedure Gtk_New (Handle : out Kernel_Handle;
Application : not null access Gtk_Application_Record'Class;
```

```
-- Pattern B: 2-space continuation Side_And_Locations : constant Message_Flags := (Editor_Side => True,
Locations => True, others => False); ``
```

4. Named Constants: Mixed_Case (Not ALL_CAPS)

Observed: Constants use Mixed_Case with underscores, not SCREAMING_SNAKE_CASE.

```
``ada -- GNAT style: Screen_Width : constant := 128; Invalid_Address : constant Address_Type := ...;
No_Breakpoint : constant := ...;
```

```
-- NOT typical GNAT style: SCREEN_WIDTH : constant := 128; -- (ALL_CAPS less common) ``
```

Note: Some embedded/hardware code uses ALL_CAPS for register constants, but Mixed_Case dominates in general GNAT code.

5. Line Width: 79 Characters

Observed: Code stays within 79 characters. Header comments use 78 characters for centered text.

```
``ada -- GNAT Studio -- -- ^-- 78 characters of dashes --^ ``
```

6. Comment Prefix: Two Spaces After - -

Observed: Consistently - - (two spaces) for regular comments.

```
``ada -- This is the standard GNAT comment format with two spaces. --  
Continuation lines maintain the same format. ``
```

7. Vertical Alignment: Widely Used

Observed: Colons, assignments, and comments aligned vertically on consecutive lines.

```
``ada -- Parameter declarations: Self : not null access constant Messages_Container'Class; Category :  
VSS.Strings.Virtual_String; File : GNATCOLL.VFS.Virtual_File  
  
-- Assignments: History_File_Base_Name : constant String := "histories.xml"; History_Max_Length : constant  
Positive := 10; ``
```

8. No Tabs

Observed: Zero tab characters across all analyzed files. Spaces only.

Generated as part of the adafmt project - an Ada 2022 linter and formatter with AI-assisted repair.